Results

Each batch is a folder which has a code that makes the data and the figures only, data can keep being reused. Each batch continues the next and so one, they shall also do the stuff needed to run all these codes to do all the analysis from the paper, so all obj use here must be created from a code like this or use the OG data. All objects or data must be made on the batches in order or come from the OG Data (if it’s not in data it must be created.) This is only for the results part of the paper.

Remember there is a methylome C1 and C2 are control, A1 and A2 are amputated. When searching for expression use the transcriptomes of the tail control and tail amputated ones. (Filters outlier, how can we find outlier and be sure)

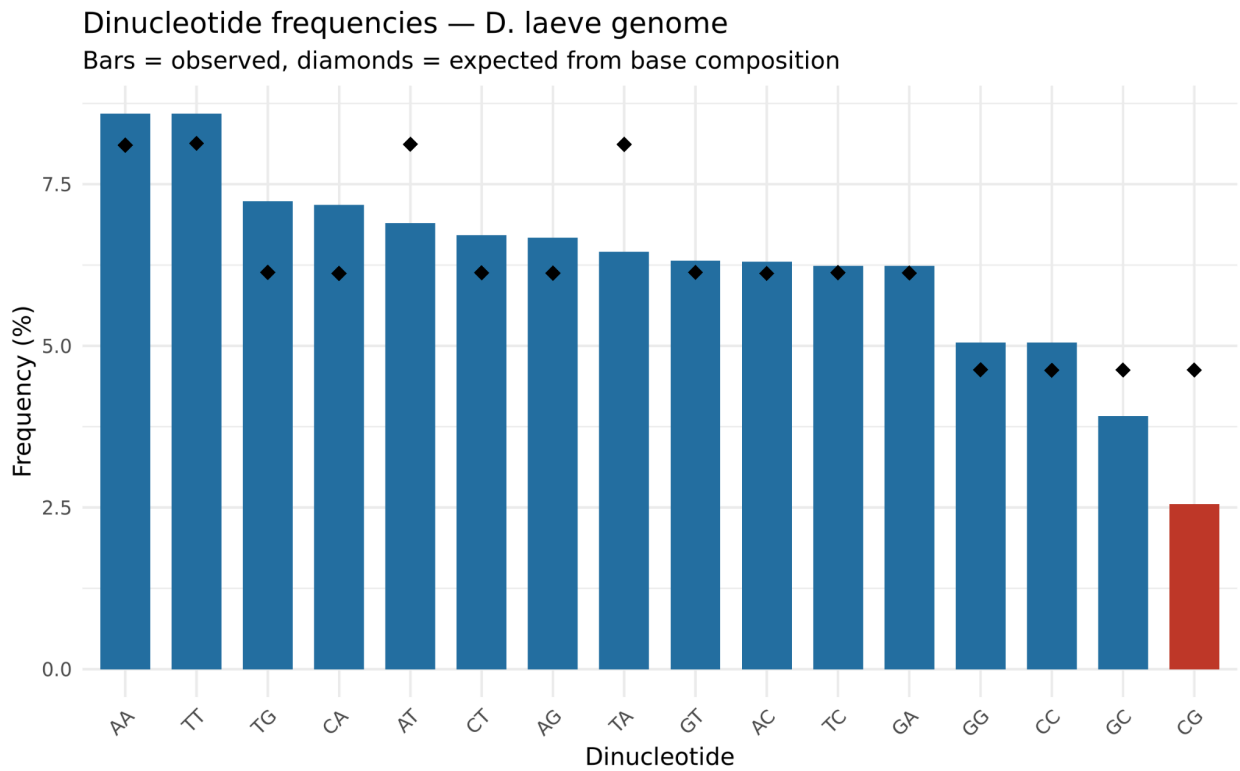
Note, collapse the methylation before measuring it.

Make a report in html by batch.

All folder have 1 or many R codes, for motif maybe pearl in homer but JASPAR in R. To have the annotation data analsys

Batch 1 (Genomic annotation of CpGs)

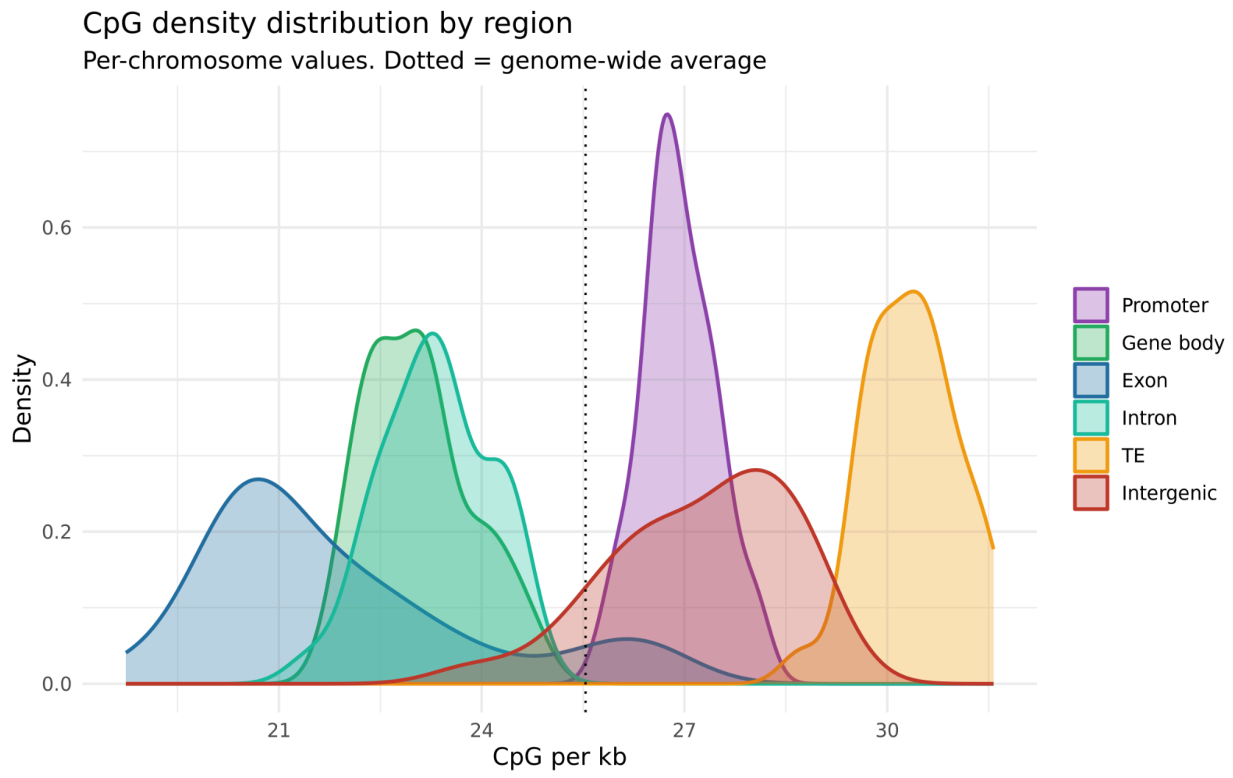
This part is done, but the plots have some outliers in the distribution that wasn’t corrected. It’s just genomic stuff related to methylation, no methylation analysis yet here.

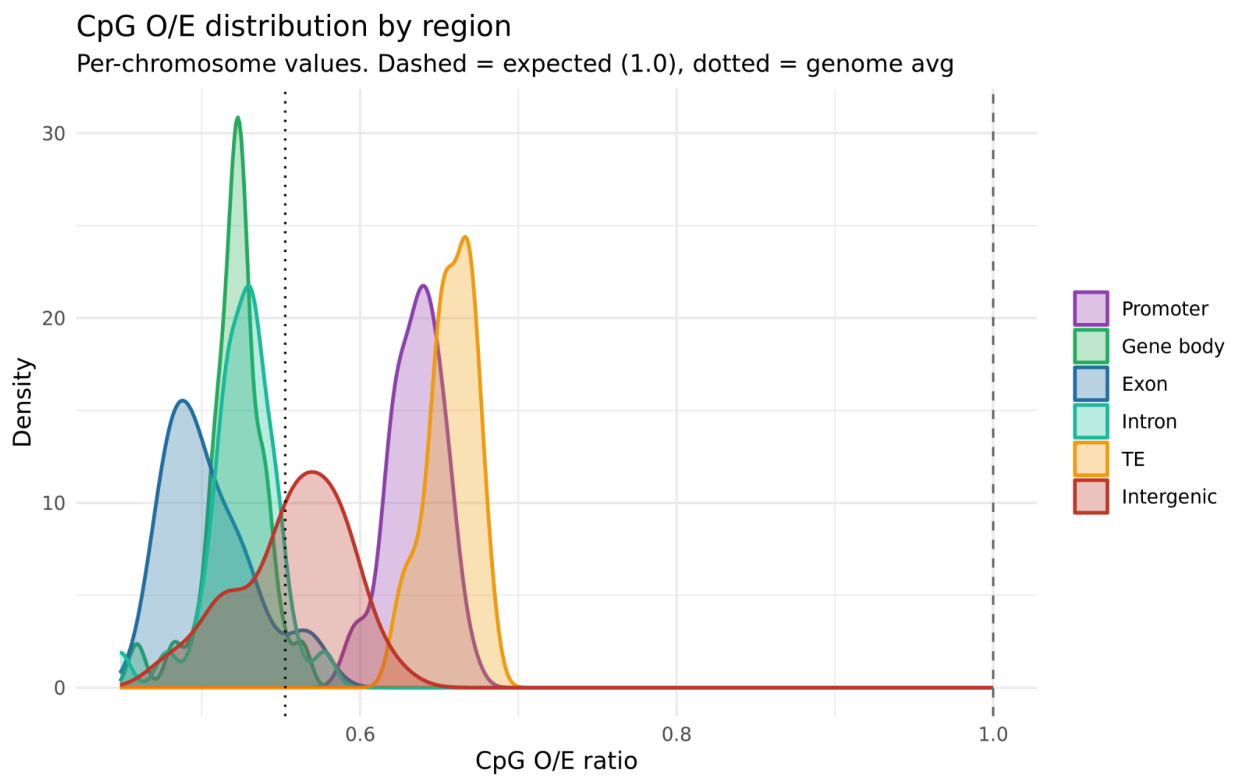


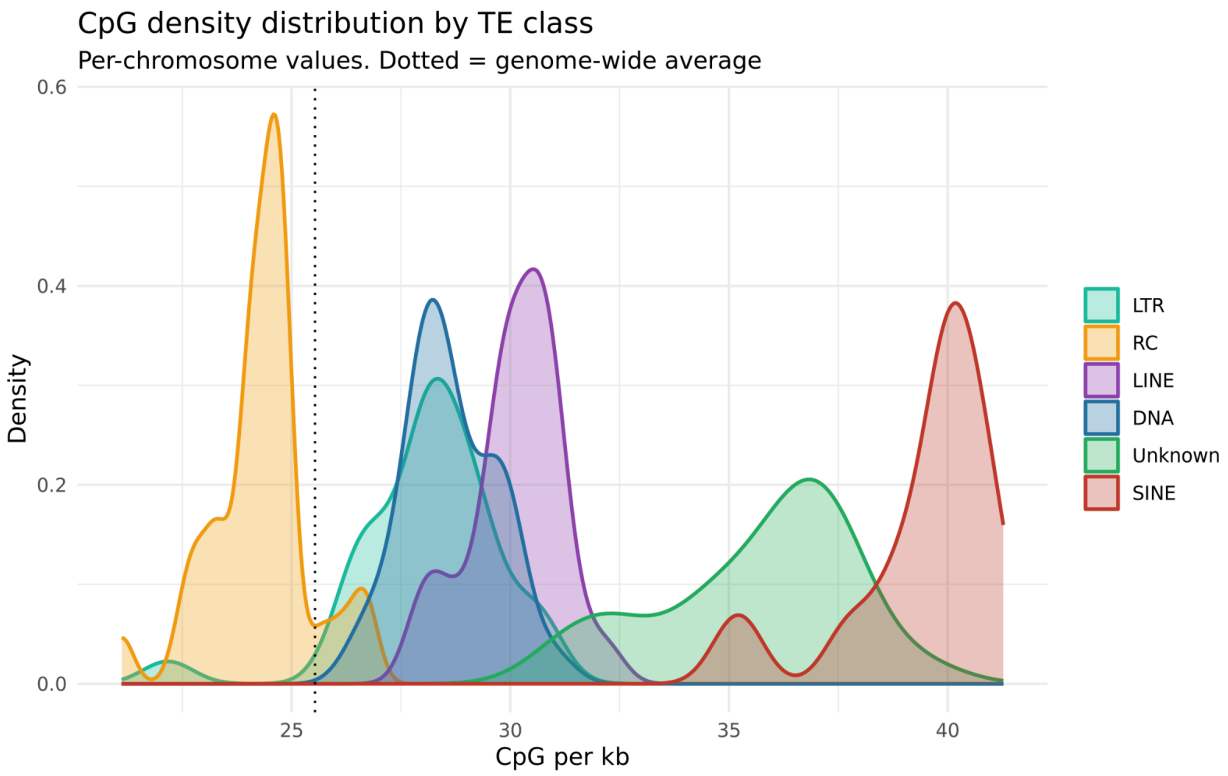
Batch 1.5

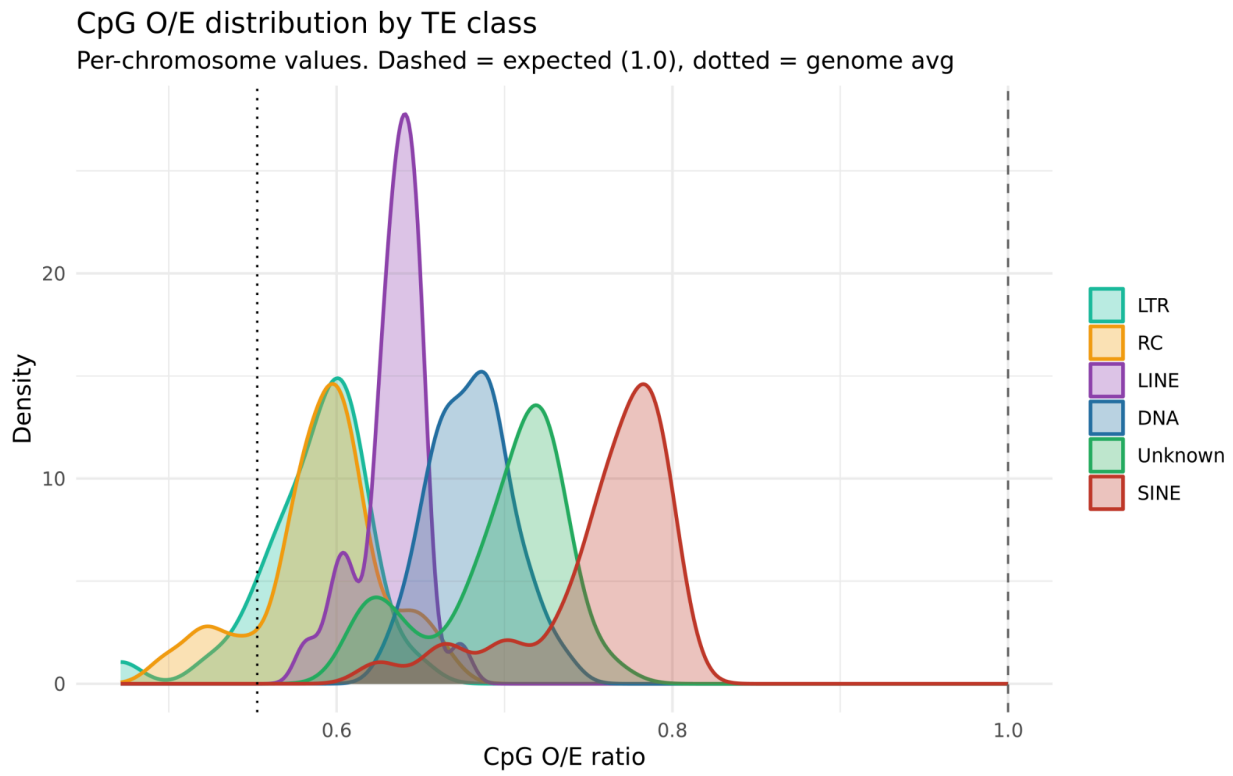
We need an in silico annotation fo Transcription factor binding sites. Add methodology for this.

Batch 2


CpG density distribution by region
Per-chromosome values. Dotted = genome-wide average
Density
0.0
0.2
0.4
0.6
21
24
27
30
CpG per kb
Promoter
Gene body
Exon
Intron
TE
Intergenic

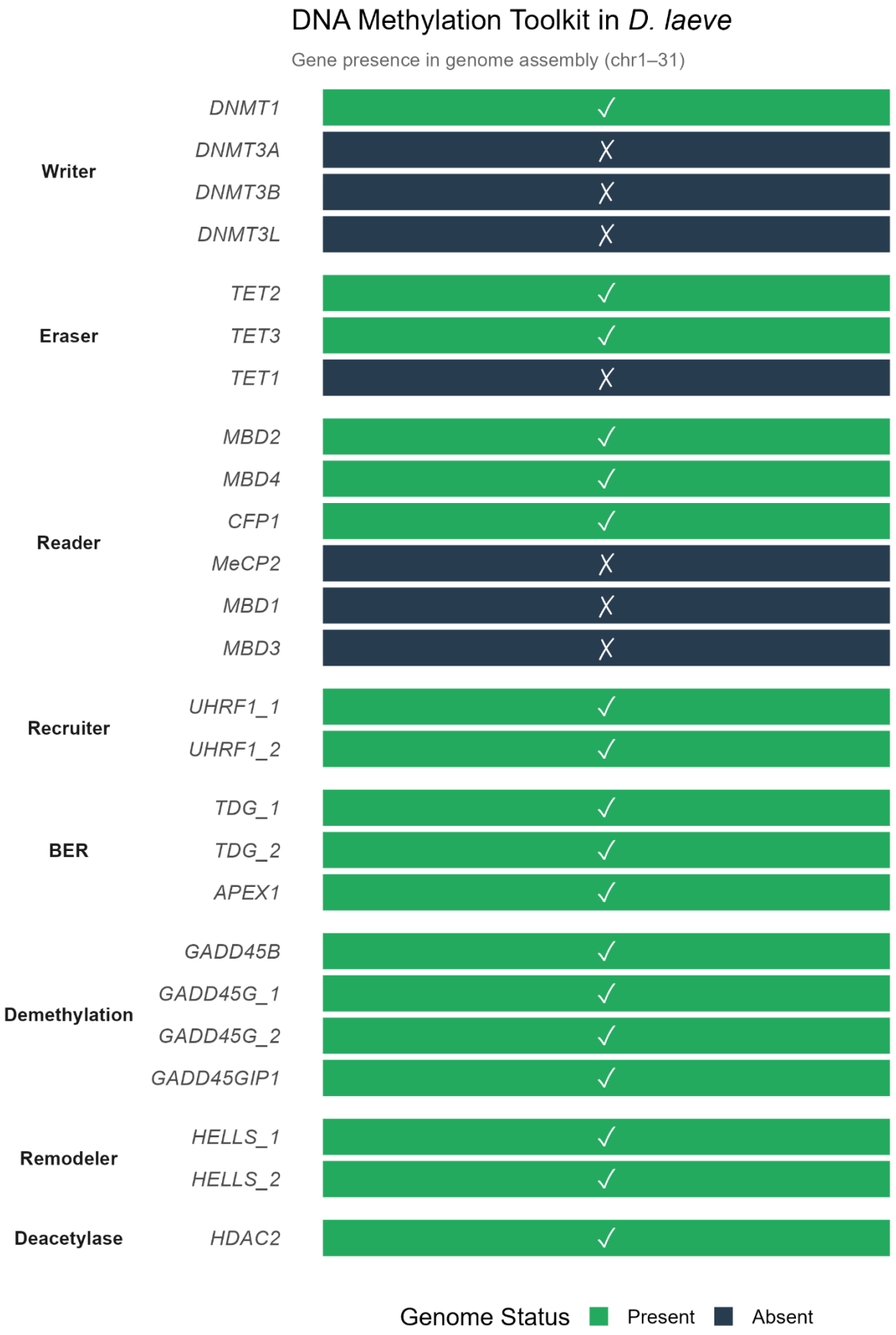

CpG O/E distribution by region
Per-chromosome values. Dashed = expected (1.0), dotted = genome avg
Density
0
10
20
30
0.6
0.8
1.0
CpG O/E ratio
Promoter
Gene body
Exon
Intron
TE
Intergenic

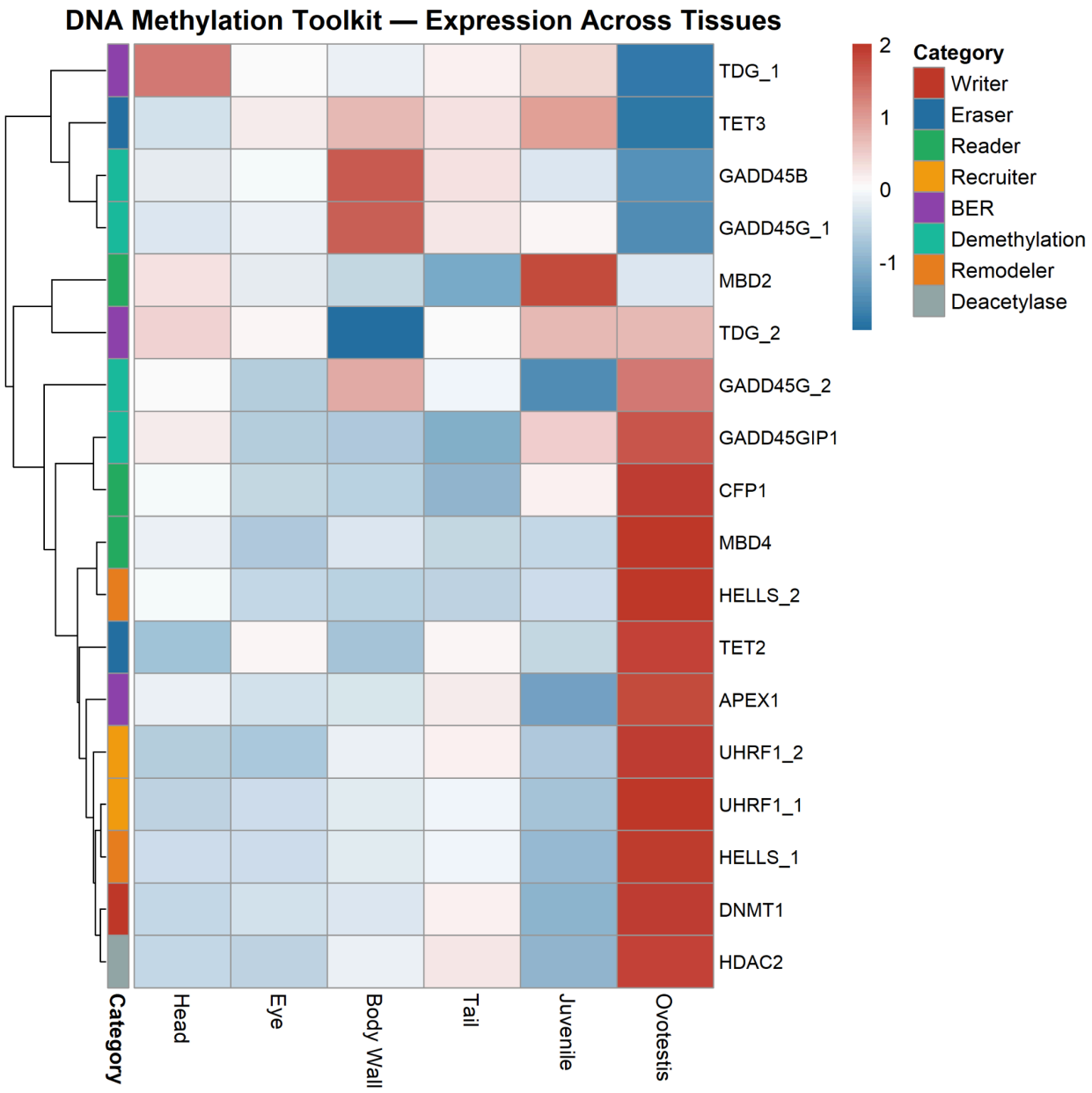

CpG density distribution by TE class
Per-chromosome values. Dotted = genome-wide average
Density
0.6
0.4
0.2
0.0
25
30
35
40
CpG per kb
LTR
RC
LINE
DNA
Unknown
SINE



Batch 2 (Methylation machinery)

Done. Methylation machinery based on the genome, and also the transcriptome by sample. Shall use the controls only and not the condition.


DNA Methylation Toolkit in D. laeve
Gene presence in genome assembly (chr1–31)
Writer
DNMT1
DNMT3A
DNMT3B
DNMT3L
Eraser
TET2
TET3
TET1
Reader
MBD2
MBD4
CFP1
MeCP2
MBD1
MBD3
Recruiter
UHRF1_1
UHRF1_2
BER
TDG_1
TDG_2
APEX1
Demethylation
GADD45B
GADD45G_1
GADD45G_2
GADD45GIP1
Remodeler
HELLS_1
HELLS_2
Deacetylase
HDAC2
Genome Status
Present
Absent



Batch 3 (Genomic CpG promoter annotation, motifs, OE + CG% in promoter (Weber), CpG Distribution vs Humans)

Needs (A) CpG O/E histogram, all promoters. (B) CpG O/E vs. GC\% scatter (Weber). (C) Class proportions. (D) Faceted histograms by class. To show that promoters are not like humans. ICP overepresented.

Only genetics

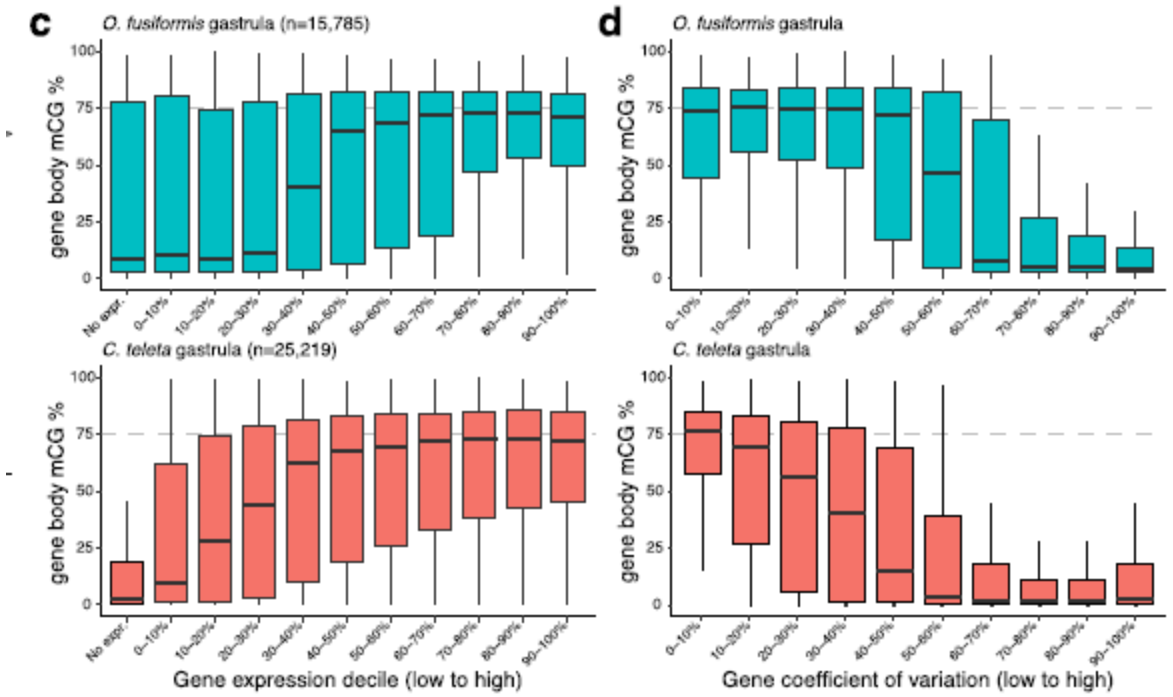
Batch 4 (Gene body methylation x Expression in decimals)

Do this to plots that we show below and also A) Gene-body meth. vs. expression. (B) Expression by methylation quartile. and by promoter, gene body and downstream.

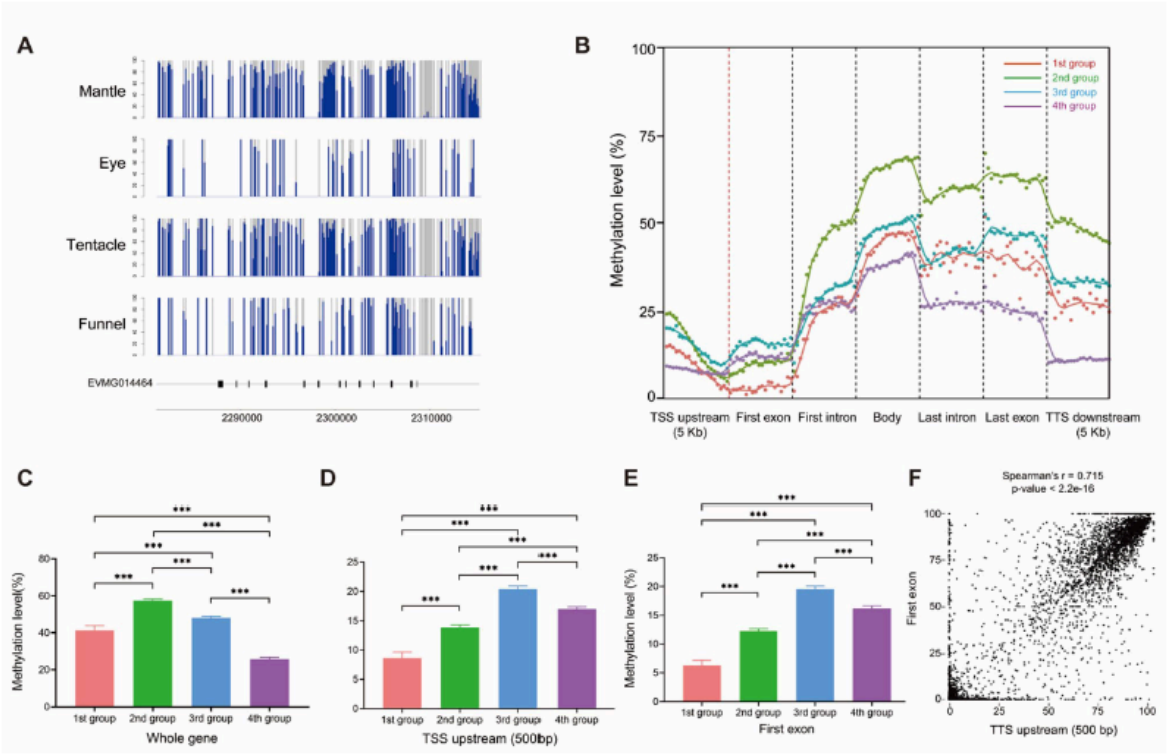
Show which regions have methylation, over or subexpected only relating to genes, overall methylation genome wide pattern. Global methylation peer sample. Some stuff of methylation tutorial should be here.

Have this, show gene body expression correlation to methylation and plots like this, see all genics. Is methylation related to expression, not differential expression here. Use promoter, exon, intron, see them individually and also as gene body. Downstream.

See this image from Annelid methylomes reveal ancestral developmental and aging-associated epigenetic erosion across Bilateria which is in papers folder



This graph is from DNA methylation landscape of Nautilus Pompilius. Must have a methylation landscape of figure B. In general, and also comparing both.



Batch 5

Here we look and Transposable elements and methylation patterns. Show that TE age is anticorrelated with methylation. Also show that there is a bimodal distribution (Ridge plots). We see that TEs show low level methylation unlike mammals (Huges findings). Kimura is better in million of years.
TEs are note very methylated (A) Kimura divergence vs. methylation by TE class. (B) Intronic vs. intergenic TE methylation (Is there any relation in TE methylation if they overlap with a gene compared over those that don’t).

Batch 6

Differential expressions come here, DMRs and DMPs.

(A) PCA. (B) Spearman heatmap. (C) Chromosome methylation. (D) DMP volcano. (E) DMR Volcano (F)DMR genomic annotation.(G) Ontology Enrichment DMP (H) Ontology enrichment DMR

Enrichment KEGG, GO, REACTOME

Batch 7

Here we try to find DMR or DMPs near genes, either inside genes(gene body) promoter, or near intergenic (enhancer silencers) to see if DMR or DMPs related to expression. We find some but without the motif stuff, and only seeing in some genes that are highly expressed 3 DMPs is not enough to support. For example DMRs near SOX19, we can only know if they are enhancer if we have the motifs, if not this part will be all diff methylation not correlated to expression
(A) Δ meth vs. Δ expression scatter. (B) DMR-gene focused. (C) Top DMR gene heatmap.

If here comes

Batch 8

Here are the modules of the GENE correlation network stuff. Some modules in WGCN are overrepresented by methylation more than change, which module is the ovotestis module. They new one even if a new color should find this if this not an artifact.

Do, enrichment of those modules, see what they are related.

Methylation Changes are associated to specific expression modules (yellow ovotestis, pluripotency, morphogenesis, cytoskeleton) or TF Gene Regulatory networks?

(A) Δ meth vs. Δ expression scatter. (B) DMR-gene focused. (C) Top DMR gene heatmap.

Batch 9

Here we see if TF, GENIE3 top nodes. If a TF has a gene, which has near or inside a motif of it we know the TF, the transcription factor and its target have not only correlational statistics, this is the GENIE3 stuff (I believe there is data for this in data if not I'll upload it, or in the cluster)

If here comes good we can add the SOX 19 and other genes which have dmr and dmp and diff expression plotted like the plots the first claude agent had, but we need to motif stuff to cement that.

Batch 10

Perplexity analysis, here is the entropy stuff, the question is does global entropy changes towards more or less when regenerating. Give plots for this that are clear

Notes:

All plots should be clear and readable for paper, check the outputs of the data, if they dont look good, or the are decentered or maybe could be plotted better for visual change it

This is supposed to be the final stuff for having the paper figures